Measuring direct and Indirect Impacts in a Multi-Sided Marketplace
Evidence from a Clustered Multiple Randomization Experiment

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1 Introduction

Direct and indirect impacts in a multi-sided marketplace In many cases experimenters may be interested in measuring impacts of their experiment within a multi-sided marketplace. In such cases, a treatment tested in an experiment may directly impact one side of the market, but also invoke responses on the other side of the market. An example is an experiment that tests a new search algorithm in an online marketplace. Algorithm changes may impact the search results displayed to customers, but may also invoke responses from sellers. Sellers may learn that their listings rank changed and make adjustments accordingly, for instance to the listed price. Throughout the paper we discuss the example of a search algorithm experiment, but similar multi-sided marketplace impacts arise in other common experiment settings, such as advertising policy changes.

A treatment that impacts units across multiple sides of the marketplace can induce complex reactions, triggering different effects: **i)** Direct Impact [D.I.] of the treatment — the "mechanical" impact of the search algorithm change on customer behavior (e.g., purchases) **ii)** Indirect Impact [I.I.] through seller adjustments, caused by their response to the algorithm change. We then decompose the total impact as follows:

$$\underbrace{\text{Total Impact}}_{\text{[T.I.]:=[D.I.]+[I.I.]}} = \underbrace{\text{Direct Impact}}_{\text{[D.I.]}} + \underbrace{\text{Indirect Impact.}}_{\text{[I.I.] - (seller response)}} \tag{1}$$

Indirect impacts [I.I.] are important as they reveal how sellers adjust their listings, such as listing prices, in response to the algorithm changing. For instance, if the algorithm used to rank products heavily prioritizes the price of goods, this may incentivize sellers to frequently update and adjust their prices in order to improve their product's ranking.

We note that 'standard" experiments that randomize at either the demand level (i.e., at the customer or search query level, exposing randomly selected customers or search queries to the treatment), or at the supply level (i.e., at the seller level, exposing randomly selected sellers to the treatment pricing scheme) are unable to recover these different impacts. In particular, experiments trying to measure the indirect response from sellers have a number of challenges:

C.1 *Demand-randomized experiments lead to mixed exposure for sellers, invalidating standard analyses*, e.g. in customer or search query randomized experiments, sellers are exposed to multiple treatments (e.g., different search algorithms) depending on the customer's treatment status.

C.2 *Seller-randomized experiments erode trust.* Seller-randomized experiments might expose different sellers to treatment within the same search page, potentially eroding their trust and willingness to use the service in the future.

C.3 *Supply-randomized experiments suffer from cross-unit interference, or spillover, invalidating causal claims under standard analyses.* Differentially treating sellers that interact with one another (e.g., within the same search page) can lead to interference, violating standard assumptions needed for the analysis of A/B tests, introducing bias in the estimates [Chawla et al., 2016, Pouget-Abadie et al., 2018].

Challenges C.1–C.3 arise when experimenters want to test the effect of interventions on sellers, but direct randomization on these units is either unfeasible (e.g., policy constraints, C.2) or can induce interference (C.3) [Holtz et al., 2024]. In these contexts, grouping similar sellers into clusters, and randomizing at the seller-cluster

level is the standard approach. Via seller-cluster randomization, experimenters avoid the problems of mixed exposure (C.1), and, if clusters capture market-competition structure, minimize bias due to interference and spillover effects (cf. C.3). For online pricing experiments, however, *direct* seller-cluster randomization is not desirable, as it can erode trust in the sellers (C.2). We employ a similar approach by clustering and randomizing at the customers' search-query level.

Our contribution Randomizing at the query-cluster level allows us to measure the *total* impact of the treatment [T.I.], but it does not allow us to separate the direct effect of the change [D.I.], from downstream indirect effects coming from sellers [I.I.]. To overcome this limitation, we develop a novel experimental design to distinguish direct from indirect effects, while retaining the desirable properties of a query-cluster randomized design, allowing to estimate both indirect and spillover effects. Our design is an instance of a multiple randomization design [MRD] [Bajari et al., 2021, Johari et al., 2022, Bajari et al., 2023, Masoero et al., 2024], a novel class of experimental designs tailored to capture cross-unit interference. We combine a simple customer-randomized experiment with a search-query cluster randomized experiment, and implement our design in a large scale online experiment. We show on real data that our approach successfully allows us to identify total, direct and indirect effects in the experiment. To the best of our knowledge, our work is the first successful large-scale implementation of this type of design.

2 Multiple randomization designs meet cluster-randomized experiments

Clustering Prior to the experiment, we form a bipartite graph $\mathcal{G} = (\mathcal{V}_{\text{se}}, \mathcal{Q}, \mathcal{E})$ linking sellers $\mathcal{V}_{\text{se}} = \{s_1, s_2, \ldots\}$ to potential search queries $\mathcal{Q} = \{q_1, q_2, \ldots\}$ issued by customers. We encode this relationship via a set of edges $\mathcal{E} \subset \mathcal{V}_{\text{se}} \times \mathcal{Q}$. We then use this graph $\mathcal{G}$ in combination with a clustering algorithm to create an exhaustive partition of the space of customers' search queries $\mathcal{Q}$ into K disjoint clusters, $\gamma_1, \ldots, \gamma_K$, with $\gamma_k \cap \gamma_{k'} = \varnothing$ for $k \neq k'$ and $\cup_{k=1}^{K} \gamma_k = \mathcal{Q}$. The clustering algorithm minimizes a notion of graph-cut, minimizing the number of clusters to which sellers are linked. Given this clustering, we let $\text{cl}(q) \in \{\gamma_1, ..., \gamma_K\}$ denote the cluster to which query $q \in \mathcal{Q}$ belongs.

cMRD and query-cluster randomization In our cMRD we do not directly randomize at the seller level, since this could erode seller trust (C.2): sellers s_1, s_2 could be exposed to different treatments within the search query. By randomizing instead at the search-query level we avoid this issue. A complication of query-cluster randomization, however, is that it induces mixed exposure for sellers: namely, sellers experience treatment for certain queries, and control for other queries. By clustering search queries and randomizing at the search-query cluster level we mitigate this issue. Conceptually, clustering creates "parallel" marketplaces, consisting of many search-query clusters, in such a way that most sellers only interact with search-queries within the same cluster and are therefore mostly exposed to the same treatment, allowing us to measure the impact of the treatment on the whole marketplace while minimizing seller mixed exposure.

Types in cMRD Via double randomization, each search-query, customer pair (q, c) belongs to one of four "types" [Bajari et al., 2023]: (a) tr: pair (q, c) exposed to the treatment ($w^{\text{Q}}_{\text{cl}(q)} = 1$ and $w^{\text{C}}_{c} = 1$). (b) ic: inconsistent customer experience, pair (q, c) exposed to control, where the same customer c would observe treatment for a search-query q' with $w^{\text{Q}}_{(q')} = 1$. (c) iq: inconsistent query experience, pair (q, c) exposed to control, where the same query q would be treated when requested by a customer c' such that $w^{\text{C}}_{c'} = 1$. (d) cc: consistent control, pair (q, c) exposed to control, where customer c always triggers control, for all search queries q', and search-query q always leads to search results where the control algorithm is applied, for any customer c'.

Direct, Indirect, Total effects By comparing types, we can estimate direct, total and indirect impacts ([D.I.], [T.I.], [I.I.]) of the intervention (cf. Equation (1)): (i) [D.I.]: the effect measured via a customer-randomized experiment, tr vs. iq (ii) [T.I.]: the effect measured via a cluster-randomized experiment, tr vs. ic (iii) [I.I.]: there are no ex ante differences between unit pairs in iq and cc, and sellers cannot respond separately to treated and control customers. Therefore, if sellers react to the treatment, their response must be applied

to all customers. Hence, we can interpret ex post differences between iq and cc as the (indirect) response of the sellers to treatment. Moreover, because we do not expect any spillover between customers (i.e., ic to be comparable to cc), the total impact [T.I.] can also be obtained by comparing tr vs. cc:

$$\underbrace{\text{tr vs. cc}}_{\text{[T.I.]}} = \underbrace{(\text{tr vs. ic})}_{\text{[T.I.]}} + \underbrace{(\text{ic vs. cc})}_{0} = \underbrace{(\text{tr vs. iq})}_{\text{[D.I.]}} + \underbrace{(\text{iq vs. cc})}_{\text{[I.I.]}}. \tag{2}$$

Simple customer- or query-cluster- randomized experiments would not allow us to isolate the indirect impact [I.I.]. cMRDs allow us to do so by comparing iq vs. cc, the *downstream* impact of the sellers' response, rather than the *intermediate* impact.

3 Empirical strategy

Each customer, query pair (c, q) has a unique treatment assignment $w_{c,q} := w_c^{\mathrm{C}} w_{\mathrm{cl}(q)}^{\mathrm{Q}} \in \{0, 1\}$. We let $y_{c,q}$ denote the total activity at the customer-query pair level (e.g., total sales revenue related to query q by customer c).

Customer-level impacts tr vs. iq and ic vs. cc correspond to simple customer-randomized experiment:

(a) tr vs. iq: Let $\mathcal{Q}_1 := \{q \in \mathcal{Q} \ : \ w_{\mathrm{cl}(q)}^{\mathrm{Q}} = 1\}$ identify queries eligible for treatment. On $\mathcal{Q}_1$, the cMRD can be viewed as a customer-randomized experiment, where pairs with type "tr" are the treated group, and pairs with type "iq" are the control group. We adopt a linear regression model with interactions, using pre-experimental measurements of the outcome variable as covariates. We let $\dot{x}_c^{\mathrm{C}} := x_c^{\mathrm{C}} - \bar{x}^{\mathrm{C}}$ be the demeaned version of x_c^{C}, where x_c^{C} is the total pre-experimental activity associated with customer c and $\bar{x}^{\mathrm{C}}$ is the average of x_c^{C} across all c. Similarly, $y_c^{\mathrm{C}} := \sum_{q \in \mathcal{Q}_1} y_{c,q}$ is the total activity associated with customer c summed across all queries the customer interacted with during the experiment. We fit via least squares $y_c^{\mathrm{C}} = \beta_0 + \beta_1 w_c^{\mathrm{C}} + \beta_2 \dot{x}_c + \beta_3 \dot{x}_c w_c^{\mathrm{C}} + \epsilon_c$ and estimate the treatment effect tr vs.iq via $\hat{\beta}_1$.

(b) ic vs. cc: Let $\mathcal{Q}_0 := \mathcal{Q} \setminus \mathcal{Q}_1$; on $\mathcal{Q}_0$, outcomes of types "ic" and "cc", outcomes should resemble an A/A test. Outcomes of customers eligible for treatment (c such that $w_c^{\mathrm{C}} = 1$) should be comparable to outcomes of customers not eligible for treatment (c such that $w_c^{\mathrm{C}} = 0$). Indeed in both cases, due to the double randomization, these pairs never receive treatment. We adopt a linear regression approach, now with $y_c := \sum_{q \in \mathcal{Q}_0} y_{c,q}$. In this case, we expect the fitted coefficient $\hat{\beta}_1$ to not be significantly different from 0.

Search query-level impacts Similar to customer-level impacts, we can estimate effects at the search query level, q, as a proxy for seller-level effects. Let $y_q := \sum_c y_{c,q}$ be total outcome related to search query q. We estimate the following regression: $y_q^{\mathrm{Q}} = \beta_0 + \beta_1 w_{\mathrm{cl}(q)}^{\mathrm{Q}} + \beta_2 \dot{x}_q + \beta_3 w_{\mathrm{cl}(q)}^{\mathrm{Q}} \dot{x}_q \epsilon_q$, We fit this model using cluster-robust standard errors to account for within-cluster correlation in the error term; this is a consequence of the clustered randomization design. β_1 is again the treatment effect that we are interested in—it quantifies ic vs. cc.

Mixed exposure and seller impacts In cMRDs, analyzing sellers' response to the treatment is complicated by mixed exposure: any seller s_i is typically exposed to both treated and control queries, i.e. treatment status (fraction of treated queries with which they interact) $t_i \in [0, 1]$ is non-binary. See [C.2, C.3] in Section 1. We estimate seller-level impacts using the following regression: $y_i^{\mathrm{A}} = \beta t_i + \theta_1 t_i \dot{x}_i^{\mathrm{A}} + \theta_2 \dot{x}_i^{\mathrm{A}} + \epsilon_i$. Here, y_i^{A} is the outcome metric for seller i, summed across all queries c the seller interacted with, and $\dot{x}_i^{\mathrm{A}}$ is a demeaned vector of covariates measured at the individual seller level. Large sellers likely have a larger difference between treated and control spending (as treatment is normalized between 0 and 1) compared to small sellers. If left unaddressed, we would therefore expect the magnitude of the estimated treatment effect estimate to be biased upwards. This bias could be more severe (due to misspecification) if the seller's response $\mathbb{E}[y_i^{\mathrm{A}} \mid t_i]$ is non-linear, and would disproportionately affect sellers with a great deal of mixed exposure, so $t_i \approx 1/2$. Directly using seller s_i's treatment share t_i in the linear model has at least two risks:

R.1 *Dependence between mixed exposure and heterogeneous treatment leads to bias.* Sellers may respond to the treatment heterogeneously in a way that varies with treatment exposure, with larger sellers more likely to suffer from worse mixed exposure.

R.2 *Non-linear seller response.* In our setting, it is plausible that the seller's response $\mathbb{E}[y_i^{\mathrm{A}} \mid t_i]$ is non-linear, resulting in bias due to misspecification of the imposed linear model, disproportionately affecting sellers with a great deal of mixed exposure, so $t_i \approx 1/2$.

To address (R.1, R.2), we extend the recent methods of Harshaw et al. [2023] and use re-randomization to obtain a "generalized propensity score" [GPS] capturing the probability that seller s_i has an observed treatment exposure $t \in [0,1]$ over hypothetical re-randomization of the cluster *and* session assignments, assuming the bipartite graph $\mathcal{G}$ is fixed. The simplest of these involves the binarized treatment t_i^{bin}. By using the GPS, we can compute for each seller a_i the binary treatment propensities $p_i^1 = \mathbb{P}(t_i^{\mathrm{bin}} = 1) = \mathbb{P}(t_i > 0.9)$ and $p_i^0 = \mathbb{P}(t_i^{\mathrm{bin}} = 0) = \mathbb{P}(t_i < 0.1)$, by simulating the treatment assignment many times. We may then compute the Hajek inverse propensity-weighted estimator

$$\hat{\beta} = \frac{\sum_i y_i^{\mathrm{A}} t_i^{\mathrm{bin}}/p_i^1}{\sum_i t_i^{\mathrm{bin}}/p_i^1} - \frac{\sum_i y_i^{\mathrm{A}}(1-t_i^{\mathrm{bin}})/p_i^0}{\sum_i (1-t_i^{\mathrm{bin}})/p_i^0}.$$

$\hat{\beta}$ is unbiased for $\mathbb{E}[y_i \mid t_i^{\mathrm{bin}} = 1] - \mathbb{E}[y_i \mid t_i^{\mathrm{bin}} = 0]$, addressing the potential dependence between t_i^{bin} and heterogeneous treatment effects (R.1). Binarizing treatment exposure and considering only units with a treatment share near 0 or 1 addresses the issue of non-linearity (R.2).

4 Experiments

Experimental setup We tested the methodology in a real-world experiment in a two-sided marketplace. While we do not disclose the exact nature of the experiment, we take the example of a search algorithm change described before to illustrate results. We implemented the cMRD by randomizing at the search query cluster and customer level. To perform clustering at the search-query level, we first constructed a weighted bipartite graph $\mathcal{G} = (\mathcal{V}_{\mathrm{se}}, \mathcal{Q}, \mathcal{E})$ consisting on a large seller graph consisting of several thousand units and hundreds of thousands of search queries. Weighted edges $\mathcal{E}$ between sellers and search-queries were based on interactions in a four-week look-back period. We use a spectral co-clustering algorithm on the graph $\mathcal{G}$, which minimizes the number of edges between clusters. These clusters were then stratified and randomized into treatment and control within each stratum. This ensures treatment and control group clusters are balanced on key metrics of interest. 5% of customers were randomized to the control holdout via a gating experiment, while 95% of customers were randomized to treatment or control according to the query-cluster treatment assignment.

Empirical results We show in Table 1 estimates for the pairwise comparisons discussed in Equation (2). The direct impact of treatment on sales was -10.67% ($p < 0.0001$). Sellers partially offset the negative direct impact of the treatment, as demonstrated by a significant 0.51% ($p = 0.011$) increase in sales through the measured indirect impact. This reflects seller adjustments in response to the treatment, including for instance changes in prices, or other offer attributes. The search query level impact of the treatment is estimated to be −10.13% ($p < 0.0001$). Last, we expected the control spillover ic vs. cc to be zero, and the empirical results confirm our expectation (0.10%, $p = 0.321$). From these pairwise comparisons, we can estimate the *total impact* [T.I.] both via (tr vs. iq) + (iq vs. cc), and (tr vs. ic) + (ic vs. cc), as per Equation (2). The estimate for the total impact obtained using the first pairwise comparison is −10.16%, while using the latter yields −10.03% which are statistically indistinguishable. In short, the total impact stemming from treatment can be decomposed in a direct impact of −10.67% and an indirect +0.51% increase through the offsetting seller response.

Comparison	Interpretation	% Impact	CI
tr vs.iq	[D.I.]	-10.67%	(-10.83%, -10.50%)
tr vs.ic	Search-query cluster impact	-10.13%	(-11.3%, -8.97%)
iq vs.cc	[I.I.]	0.51%	(0.1%, 0.9%)
ic vs.cc	Control Spillover	0.10%	(-0.09%, 0.29%)

Table 1: Average percent impact estimates from the cMRD.

Moreover, we obtained a seller-level comparison using the bipartite-weighted approach introduced in Section 3, obtaining similar results. For example, we estimated a -10.3% statistically significant impact ($p < 0.0001$, CI: $[-12.63\%, -7.98\%]$) using the exposure weighted model.

5 Conclusion and next steps

We proposed and implemented a new experimental design, the cMRD, which combines a cluster-randomized experiment at the search-query level with a customer-randomized experiment. We showed both in theory and in a real-world experiment that cMRDs improve upon standard experiments by allowing us to precisely estimate the differential impacts of a treatment on sellers and customers, and to separate the direct, customer-level impact from the indirect impact through seller response. Future workstreams include:

FW.1 *Improving imperfect clustering.* Clustering might not perfectly capture the relationship between sellers and search queries. Even under complete knowledge of the graph, it is unlikely to completely eliminate spillover effects when strong network interference is present. We plan to develop methods to assess analysis robustness to uncertainty in the graph structure, further enhancing bias correction due to interference.

FW.2 *Robustness to extrapolation when treatment assignments are diluted.* Due to customer-level randomization, fewer sellers are administered near 100% treatment exposure. Our analysis relies on linear extrapolation for treatment effect estimation, hence is potentially not robust to non-linear seller response. We are conducting simulations to assess the robustness of the proposed approaches to different patterns of heterogeneous treatment effects and non-linear seller responses, in order to improve our estimates.

FW.3 *Adoption in new experimental settings.* We plan to use this design also in other settings (e.g., measure the downstream impact of an advertiser-facing change on customers or a creator-facing change on viewers).

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